

12.1 Intervals for estimable functions

A point estimate needs a statement of its precision. A confidence interval gives one exactly: a range computed from the data that contains the unknown quantity with a probability fixed in advance. This section builds the interval for one estimable function, relates it to the t test, and records two facts that are easy to overlook: an interval for a non-estimable function is useless, and the interval for σ^2 , unlike the one for a coefficient, depends delicately on normality.

Throughout the chapter we work in the normal linear model

$$\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2 \mathbf{I}), \quad \boldsymbol{\beta} \in \mathbb{R}^p, \sigma^2 > 0,$$

of Eq. 7.3.1, with $r = \text{rank}(\mathbf{X}) < n$. We write $\mathbf{G}$ for any generalized inverse of $\mathbf{X}^\top \mathbf{X}$, $\hat{\boldsymbol{\beta}}$ for any least squares estimate, $\text{SSE} = \|(\mathbf{I} - \mathbf{M})\mathbf{Y}\|^2$ and $s^2 = \text{SSE}/(n - r)$. For $0 < \gamma < 1$, $t_{\nu, \gamma}$, $F_\gamma(q, \nu)$ and $\chi_\gamma^2(\nu)$ denote the upper γ points of $t(\nu)$, $F(q, \nu)$ and $\chi^2(\nu)$.

12.1.1 Confidence sets

Definition 12.1.1 — Confidence set

Let $\theta = g(\boldsymbol{\beta}, \sigma^2)$ be a quantity with values in a set Θ . A **confidence set** for θ with **confidence level** $1 - \alpha$ is a rule that assigns to each data vector $\mathbf{y}$ a subset $C(\mathbf{y}) \subseteq \Theta$, such that

$$\Pr_{\boldsymbol{\beta}, \sigma^2} \{\theta \in C(\mathbf{Y})\} \geq 1 - \alpha$$

for every $\boldsymbol{\beta} \in \mathbb{R}^p$ and $\sigma^2 > 0$. The left side is the **coverage probability** at $(\boldsymbol{\beta}, \sigma^2)$. The set is **exact** if the coverage equals $1 - \alpha$ at every parameter value. When $C(\mathbf{y})$ is an interval it is a **confidence interval**.

The probability is taken over $\mathbf{Y}$ with the parameters fixed: the set is random and the target is not. A realized interval either contains θ or it does not; $1 - \alpha$ describes the procedure.

The standard way to build an exact set is a **pivot**: a function $Q(\mathbf{Y}, \theta)$ whose distribution is the same at every parameter value. If $\Pr\{Q(\mathbf{Y}, \theta) \in A\} = 1 - \alpha$ for a fixed set A , then $C(\mathbf{y}) = \{\theta_0 : Q(\mathbf{y}, \theta_0) \in A\}$ is exact, because the events $\{\theta \in C(\mathbf{Y})\}$ and $\{Q(\mathbf{Y}, \theta) \in A\}$ are the same event. All the sets of this chapter are built this way, from pivots supplied by [Chapter 7](#).

12.1.2 The t interval

Theorem 12.1.1 — The t interval for an estimable function

Let $\boldsymbol{\lambda} \in \mathcal{C}(\mathbf{X}^\top)$, $\boldsymbol{\lambda} \neq \mathbf{0}$, and put $\text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}) = s\sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$. Let

$$I(\mathbf{Y}) = [\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - t_{n-r, \alpha/2} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}}), \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + t_{n-r, \alpha/2} \text{se}(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}})]. \quad (12.1.1)$$

- (a) $I(\mathbf{Y})$ does not depend on the choice of $\hat{\boldsymbol{\beta}}$ or of $\mathbf{G}$.
- (b) $I(\mathbf{Y})$ is an exact $1 - \alpha$ confidence interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$.

- (c) Its length $L = 2t_{n-r,\alpha/2} s \sqrt{\lambda^\top \mathbf{G} \lambda}$ has expectation $E L = 2t_{n-r,\alpha/2} \kappa_{n-r} \sigma \sqrt{\lambda^\top \mathbf{G} \lambda}$, where

$$\kappa_\nu = E \sqrt{V/\nu} = \sqrt{\frac{2}{\nu}} \frac{\Gamma((\nu+1)/2)}{\Gamma(\nu/2)} < 1, \quad V \sim \chi^2(\nu).$$

- (d) The one-sided intervals $(-\infty, \lambda^\top \hat{\beta} + t_{n-r,\alpha} \text{se}]$ and $[\lambda^\top \hat{\beta} - t_{n-r,\alpha} \text{se}, \infty)$ are exact $1 - \alpha$ confidence intervals for $\lambda^\top \beta$.

Proof.

- (a) Write $\lambda = \mathbf{X}^\top \rho$. Then $\lambda^\top \hat{\beta} = \rho^\top \mathbf{X} \hat{\beta} = \rho^\top \mathbf{M} \mathbf{Y}$ and $\lambda^\top \mathbf{G} \lambda = \rho^\top \mathbf{X} \mathbf{G} \mathbf{X}^\top \rho = \rho^\top \mathbf{M} \rho$ by Theorem 6.3.7. Neither involves the choices, and neither does s .
- (b) By Corollary 7.5.2, $T = (\lambda^\top \hat{\beta} - \lambda^\top \beta) / \text{se}(\lambda^\top \hat{\beta}) \sim t(n-r)$ at every (β, σ^2) . The event $\lambda^\top \beta \in I(\mathbf{Y})$ is the event $|T| \leq t_{n-r,\alpha/2}$, whose probability is $1 - \alpha$. So part (b) restates Eq. 7.5.1 as a confidence statement.
- (c) L is a constant multiple of $s = \sigma \sqrt{V/(n-r)}$ with $V = \text{SSE}/\sigma^2 \sim \chi^2(n-r)$ (Theorem 7.5.1(b)). With the $\chi^2(\nu)$ density of Eq. 4.1.1,

$$E \sqrt{V} = \int_0^\infty \frac{v^{1/2} v^{\nu/2-1} e^{-v/2}}{2^{\nu/2} \Gamma(\nu/2)} dv = \frac{2^{(\nu+1)/2} \Gamma((\nu+1)/2)}{2^{\nu/2} \Gamma(\nu/2)},$$

because the integrand is a multiple of the $\chi^2(\nu+1)$ density. Dividing by $\sqrt{\nu}$ gives κ_ν . By Jensen's inequality $E \sqrt{V/\nu} < \sqrt{E(V/\nu)} = 1$, with strict inequality since V is not constant.

- (d) The events are $\{T \geq -t_{n-r,\alpha}\}$ and $\{T \leq t_{n-r,\alpha}\}$, each of probability $1 - \alpha$. ■

Part (a) matters when $\mathbf{X}$ is rank deficient: the interval is a property of the model, not of the software's choice of generalized inverse. The degrees of freedom are $n - r$, not $n - p$.

Part (c) shows what the interval costs. Its expected length is proportional to $\sigma \sqrt{\lambda^\top \mathbf{G} \lambda}$, the standard deviation of the estimate. Estimating σ adds the factor $t_{n-r,\alpha/2} \kappa_{n-r}$ in place of the normal point $z_{\alpha/2}$ that would be used if σ were known. For $\alpha = 0.05$ that factor is 2.45 at $n - r = 5$ and 2.06 at $n - r = 20$, against 1.96. Beyond about twenty residual degrees of freedom, precision improves only through $\lambda^\top \mathbf{G} \lambda$, that is, through the design.

Example. (Murder rates: coefficients and a mean)

Return to the regression of the 2009 murder rate of the 50 states on the poverty rate, the percentage of single-parent households and the percentage urban, with an intercept (Example (Poverty and murder rates)). Here $n = 50$, $r = 4$, $s^2 = 2.217$, and $t_{46,0.025} = 2.013$. The 95% intervals for the coefficients are

Coefficient	Estimate	Standard error	95% interval
intercept	−9.0946	1.5555	(−12.226, −5.964)
poverty	0.2538	0.0934	(0.066, 0.442)
single parent	0.3980	0.0829	(0.231, 0.565)
urban	0.0046	0.0131	(−0.022, 0.031)

The data are consistent with anything from 0.066 to 0.442 murders per 100,000 for each point of poverty, with the other regressors held fixed in the sense of Theorem 6.6.2. The urban interval straddles zero but is narrow, which says more than “not significant”: any such association is at most 0.031 per percentage point.

A mean response is an estimable function too. For a hypothetical state with poverty 18%, single-parent households 28% and 70% urban, $\mathbf{x}_0 = (1, 18, 28, 70)^\top$, the estimated mean murder rate is 6.937 with standard error 0.412, and the 95% interval is (6.107, 7.767). Here $\mathbf{x}_0^\top \mathbf{G} \mathbf{x}_0 = 0.0767$. Section 12.3 interprets this number as the leverage of the new point.

```
import numpy as np
import statsmodels.api as sm
from scipy import stats

data = sm.datasets.statcrime.load_pandas().data
data = data.drop(index="District of Columbia")
y = data["murder"].to_numpy()
X = np.column_stack([np.ones(len(y)), data[["poverty", "single", "urban"]]])
n, p = X.shape
r = np.linalg.matrix_rank(X)
G = np.linalg.pinv(X.T @ X)                    # any generalized inverse will do
beta_hat = G @ X.T @ y
s2 = np.sum((y - X @ beta_hat) ** 2) / (n - r)
q = stats.t.ppf(0.975, n - r)                  # upper 2.5% point of t(n - r)

def t_interval(lam):
    """95% interval for the estimable function lam^T beta."""
    est = lam @ beta_hat
    se = np.sqrt(s2 * lam @ G @ lam)
    return est, se, est - q * se, est + q * se

for j, name in enumerate(["intercept", "poverty", "single", "urban"]):
    est, se, lo, hi = t_interval(np.eye(p)[j])
    print(f"{name:9s} {est:8.4f} se {se:.4f} 95% interval ({lo:.3f}, {hi:.3f})")
```

[Run this code online](#)

12.1.3 What the standard error measures

For a coefficient in a full-rank model with an intercept, Eq. 6.6.2 gives

$$\text{se}(\hat{\beta}_j) = \frac{s}{\|(\mathbf{I} - \mathbf{M}_{(j)})\mathbf{x}_j\|} = \frac{s}{\sqrt{S_{jj}(1 - R_j^2)}},$$

where $\mathbf{M}_{(j)}$ projects onto the span of the other columns. The interval is short when $\mathbf{x}_j$ has a large component orthogonal to the other regressors; collinearity lengthens it through $1 - R_j^2$.

In general $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = \|\mathbf{M} \boldsymbol{\rho}\|^2$, with $\boldsymbol{\lambda} = \mathbf{X}^\top \boldsymbol{\rho}$, is the smallest squared length of a vector $\mathbf{a}$ with $\mathbf{E}(\mathbf{a}^\top \mathbf{Y}) = \boldsymbol{\lambda}^\top \boldsymbol{\beta}$ (Theorem 7.2.1).

12.1.4 Duality with tests

An interval and a family of tests carry the same information, in any model.

Proposition 12.1.2 — Tests and confidence sets

For each $\theta_0 \in \Theta$, let $A(\theta_0)$ be the acceptance region of a test of $H : \theta = \theta_0$ with $\Pr_{\beta, \sigma^2}\{\mathbf{Y} \in A(\theta_0)\} \geq 1 - \alpha$ whenever $g(\beta, \sigma^2) = \theta_0$. Then

$$C(\mathbf{y}) = \{\theta_0 \in \Theta : \mathbf{y} \in A(\theta_0)\}$$

is a $1 - \alpha$ confidence set for θ , and it is exact if every test has size exactly α at every parameter value in its hypothesis. Conversely, if C is a $1 - \alpha$ confidence set, the rule “reject $H : \theta = \theta_0$ iff $\theta_0 \notin C(\mathbf{y})$ ” is a level- α test for every θ_0 .

Proof. For every parameter value, with $\theta = g(\beta, \sigma^2)$, the events $\{\theta \in C(\mathbf{Y})\}$ and $\{\mathbf{Y} \in A(\theta)\}$ coincide, so they have the same probability. Both directions follow. ■

For an estimable function, the two-sided t test of $\boldsymbol{\lambda}^\top \beta = d$ accepts iff $|\boldsymbol{\lambda}^\top \hat{\beta} - d| \leq t_{n-r, \alpha/2} \text{se}$, which is the condition $d \in I(\mathbf{y})$. So the interval Eq. 12.1.1 is exactly the set of values d that the t test of Theorem 11.6.1 does not reject. The p -value of the test of d is the smallest α for which the $1 - \alpha$ interval excludes d .

The interval is the more informative report. A test of $\beta_j = 0$ reports one bit, confounded with the sample size; the interval shows location and resolution. Exercise (B2) uses it to answer “is the effect negligible?”, which a test of $\beta_j = 0$ cannot answer at all.

12.1.5 Functions that are not estimable

What if $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$? The recipe above fails because $\boldsymbol{\lambda}^\top \hat{\beta}$ depends on the choice of $\hat{\beta}$. The failure is not a defect of the recipe. No procedure can do better.

Proposition 12.1.3 — No useful interval for a non-estimable function

Let $\boldsymbol{\lambda} \notin \mathcal{C}(\mathbf{X}^\top)$ and let C be a $1 - \alpha$ confidence set for $\boldsymbol{\lambda}^\top \beta$ with $\alpha < 1$. Then at every parameter value (β, σ^2) and for every real v , $\Pr_{\beta, \sigma^2}\{v \in C(\mathbf{Y})\} \geq 1 - \alpha$. If the sets $C(\mathbf{y})$ are such that $\{(\mathbf{y}, v) : v \in C(\mathbf{y})\}$ is measurable, the expected length of $C(\mathbf{Y})$ is infinite at every parameter value.

Proof. By Theorem 8.3.1 there is $\mathbf{w} \in \mathcal{N}(\mathbf{X})$ with $\boldsymbol{\lambda}^\top \mathbf{w} \neq 0$. Fix (β, σ^2) and v , and put $\beta' = \beta + c\mathbf{w}$ with $c = (v - \boldsymbol{\lambda}^\top \beta) / \boldsymbol{\lambda}^\top \mathbf{w}$. Then $\mathbf{X}\beta' = \mathbf{X}\beta$, so $\mathbf{Y}$ has the same distribution under (β', σ^2) as under (β, σ^2) , while $\boldsymbol{\lambda}^\top \beta' = v$. Hence $\Pr_{\beta, \sigma^2}\{v \in C(\mathbf{Y})\} = \Pr_{\beta', \sigma^2}\{\boldsymbol{\lambda}^\top \beta' \in C(\mathbf{Y})\} \geq 1 - \alpha$. For the length, Fubini’s theorem gives, for

every $a > 0$,

$$E[\text{length}(C(\mathbf{Y}) \cap [-a, a])] = \int_{-a}^a \Pr\{v \in C(\mathbf{Y})\} dv \geq 2a(1 - \alpha),$$

and the left side is at most the expected length of $C(\mathbf{Y})$. Let $a \rightarrow \infty$. ■

Thus the algebraic fact of [Chapter 8](#) becomes a statement about inference.

12.1.6 An interval for σ^2

Proposition 12.1.4 — Interval for the error variance

The interval

$$\left[\frac{\text{SSE}}{\chi_{\alpha/2}^2(n-r)}, \frac{\text{SSE}}{\chi_{1-\alpha/2}^2(n-r)} \right]$$

is an exact $1 - \alpha$ confidence interval for σ^2 , and the square roots of its endpoints give one for σ .

Proof. $\text{SSE}/\sigma^2 \sim \chi^2(n-r)$ at every parameter value (Theorem 7.5.1(b)), so it is a pivot. The event $\chi_{1-\alpha/2}^2(n-r) \leq \text{SSE}/\sigma^2 \leq \chi_{\alpha/2}^2(n-r)$ has probability $1 - \alpha$, and it is the event that σ^2 lies in the interval, which is the interval of Exercise (B3). Taking square roots is monotone. ■

For the murder-rate regression, $\text{SSE} = 46s^2$ with $s^2 = 2.217$, the chi-squared points are 29.160 and 66.617, and the 95% interval for σ^2 is (1.531, 3.497). The interval is not symmetric about s^2 , and it is not the shortest one with this coverage ([Exercise \(C2\)](#)).

The two pivots of this section react differently to non-normal errors. The t statistic needs normality of a weighted average of errors, which a central limit theorem restores, and $s \rightarrow \sigma$, which holds for any error law. The chi-squared pivot needs the whole error distribution, because the variance of SSE depends on the fourth moment (Proposition 7.5.4). Take the fitted murder-rate model as the truth, with $\sigma^2 = s^2$, and simulate 40000 data sets. With normal errors the two 95% intervals cover in proportions 0.9514 (poverty coefficient) and 0.9491 (σ^2). With errors from a $t(5)$ distribution rescaled to the same variance, the coefficient interval still covers 0.9498 of the time, but the interval for σ^2 covers only 0.8165. More data do not repair this (Exercise (B3)).

```
x0 = np.array([1.0, 18.0, 28.0, 70.0])           # a state with high poverty and
single_parenthood
est0, se0, lo0, hi0 = t_interval(x0)
print(f"mean murder rate at x0: {est0:.3f}, se {se0:.3f}, interval ({lo0:.3f}, {hi0:.3f})")

sse = s2 * (n - r)
lo_s2 = sse / stats.chi2.ppf(0.975, n - r)      # interval for sigma^2 from SSE/
sigma^2 ~ chi^2(n-r)
```

```
hi_s2 = sse / stats.chi2.ppf(0.025, n - r)
print(f"s^2 = {s2:.3f}; 95% interval for sigma^2 ({lo_s2:.3f}, {hi_s2:.3f})")
```

[Run this code online](#)

12.1.7 Exercises

12.1.7.1 A. Check your understanding

Exercise A1.

A design is replicated m times: $\mathbf{X}$ is replaced by m stacked copies of itself. For a fixed estimable $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$, how does $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ change? Roughly how many replicates are needed to halve the expected length of the 95% interval when $n - r$ is already large, and why is the answer slightly less than 4 when $n - r$ is small?

Solution. The stacked matrix has $\mathbf{X}^\top \mathbf{X}$ replaced by $m\mathbf{X}^\top \mathbf{X}$, and $\mathbf{G}/m$ is a generalized inverse of $m\mathbf{X}^\top \mathbf{X}$, so $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}$ is divided by m . By Theorem 12.1.1(c) the expected length is proportional to $t_{\nu, \alpha/2} \kappa_\nu / \sqrt{m}$, with $\nu = mn - r$. For large ν the factor $t\kappa$ is nearly constant and $m = 4$ halves the length; for small ν , replication also lowers $t\kappa$, so slightly less suffices.

12.1.7.2 B. Practice

Exercise B1.

Show that $\kappa_\nu \rightarrow 1$ as $\nu \rightarrow \infty$, and that s/κ_{n-r} is an unbiased estimator of σ . Compute κ_ν for $\nu = 1$ and $\nu = 2$.

Solution. $V/\nu \rightarrow 1$ in probability by the law of large numbers, and $\sqrt{V/\nu}$ is bounded in mean square, hence uniformly integrable, so $E \sqrt{V/\nu} \rightarrow 1$. Unbiasedness: $E s = \sigma E \sqrt{V/(n-r)} = \sigma \kappa_{n-r}$. For $\nu = 1$, $\kappa_1 = \sqrt{2} \Gamma(1)/\Gamma(1/2) = \sqrt{2/\pi} \approx 0.798$. For $\nu = 2$, $\kappa_2 = \Gamma(3/2)/\Gamma(1) = \sqrt{\pi}/2 \approx 0.886$.

Exercise B2.

Let $\delta > 0$ be a margin below which $|\boldsymbol{\lambda}^\top \boldsymbol{\beta}|$ is considered negligible. The *two one-sided tests* procedure declares equivalence when both $H_1 : \boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq -\delta$ and $H_2 : \boldsymbol{\lambda}^\top \boldsymbol{\beta} \geq \delta$ are rejected by one-sided t tests at level α . Show that this happens iff the $1 - 2\alpha$ interval Eq. 12.1.1 lies inside $(-\delta, \delta)$. Show that the procedure has level α : when $|\boldsymbol{\lambda}^\top \boldsymbol{\beta}| \geq \delta$, the probability of declaring equivalence is at most α .

Solution. H_1 is rejected iff $(\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + \delta)/se > t_{n-r, \alpha}$, that is iff the lower end of the $1 - 2\alpha$ interval, $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - t_{n-r, \alpha} se$, exceeds $-\delta$. Symmetrically H_2 is rejected iff the upper end is below δ . For the level: if $\boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq -\delta$, declaring equivalence requires rejecting H_1 . Write $\tau = \sigma \sqrt{\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda}}$, $Z = (\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - \boldsymbol{\lambda}^\top \boldsymbol{\beta})/\tau \sim N(0, 1)$ and $V = SSE/\sigma^2 \sim \chi^2(\nu)$, $\nu = n - r$, independent of Z (Theorem 7.5.1). The test statistic is $(Z + \eta)/\sqrt{V/\nu}$ with $\eta = (\boldsymbol{\lambda}^\top \boldsymbol{\beta} + \delta)/\tau$. For every realization of (Z, V) it increases in η , so the probability that

it exceeds $t_{\nu, \alpha}$ increases in η . At $\eta = 0$ that probability is α , so it is at most α whenever $\boldsymbol{\lambda}^\top \boldsymbol{\beta} \leq -\delta$. The case $\boldsymbol{\lambda}^\top \boldsymbol{\beta} \geq \delta$ is symmetric.

Exercise B3.

Suppose $\mathbf{Y} \sim N_n(\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{V})$ with $\mathbf{V}$ known and positive definite and $\mathbf{X}$ of full column rank. Using the generalized least squares estimator $\tilde{\boldsymbol{\beta}} = (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \mathbf{X}^\top \mathbf{V}^{-1} \mathbf{Y}$ and $\tilde{s}^2 = (\mathbf{Y} - \mathbf{X}\tilde{\boldsymbol{\beta}})^\top \mathbf{V}^{-1} (\mathbf{Y} - \mathbf{X}\tilde{\boldsymbol{\beta}}) / (n - p)$, derive an exact $1 - \alpha$ interval for $\boldsymbol{\lambda}^\top \boldsymbol{\beta}$. Write it out for a straight line $EY_i = \beta_0 + \beta_1 x_i$ with $\mathbf{V} = \text{diag}(1/w_i)$.

Solution. Let $\mathbf{V} = \mathbf{L}\mathbf{L}^\top$ (Cholesky) and $\mathbf{Z} = \mathbf{L}^{-1}\mathbf{Y} \sim N_n(\mathbf{L}^{-1}\mathbf{X}\boldsymbol{\beta}, \sigma^2\mathbf{I})$. The ordinary least squares estimate for $(\mathbf{Z}, \mathbf{L}^{-1}\mathbf{X})$ is $\tilde{\boldsymbol{\beta}}$, and its residual sum of squares is $(n-p)\tilde{s}^2$. Applying [Theorem 12.1.1](#) to the transformed model gives $\boldsymbol{\lambda}^\top \tilde{\boldsymbol{\beta}} \pm t_{n-p, \alpha/2} \tilde{s} \sqrt{\boldsymbol{\lambda}^\top (\mathbf{X}^\top \mathbf{V}^{-1} \mathbf{X})^{-1} \boldsymbol{\lambda}}$. For the weighted straight line, with $\bar{x}_w = \sum w_i x_i / \sum w_i$, $\bar{y}_w$ defined likewise and $S_w = \sum w_i (x_i - \bar{x}_w)^2$, the slope estimate is $\tilde{\beta}_1 = \sum w_i (x_i - \bar{x}_w)(y_i - \bar{y}_w) / S_w$ and its interval is $\tilde{\beta}_1 \pm t_{n-2, \alpha/2} \tilde{s} / \sqrt{S_w}$.

Exercise B4.

In a one-way layout with a groups of sizes $n_1, \dots, n_a$ and the overparameterized model $EY_{ij} = \mu + \alpha_i$, show that the interval for $\alpha_1 - \alpha_2$ is $\bar{y}_1 - \bar{y}_2 \pm t_{n-a, \alpha/2} s \sqrt{1/n_1 + 1/n_2}$, with $n = \sum n_i$. What does [Proposition 12.1.3](#) say about an interval for α_1 alone?

Solution. The rank is $r = a$. With $\boldsymbol{\lambda} = \mathbf{e}_2 - \mathbf{e}_3$ (coordinates $\mu, \alpha_1, \dots, \alpha_a$), $\boldsymbol{\lambda}^\top \boldsymbol{\beta} = \alpha_1 - \alpha_2$ is estimable, and $\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} = \bar{y}_1 - \bar{y}_2$ for every least squares estimate. Its variance is $\sigma^2(1/n_1 + 1/n_2)$ since the group means are independent, so $\boldsymbol{\lambda}^\top \mathbf{G} \boldsymbol{\lambda} = 1/n_1 + 1/n_2$. Substitute in [Eq. 12.1.1](#) with $n - r = n - a$. The function α_1 is not estimable (adding c to every α_i and subtracting it from μ leaves $\mathbf{X}\boldsymbol{\beta}$ unchanged), so every 95% confidence set for it covers every real number with probability at least 0.95.

12.1.7.3 C. Going deeper

Exercise C1.

Among intervals $[\boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} - a \text{ se}, \boldsymbol{\lambda}^\top \hat{\boldsymbol{\beta}} + b \text{ se}]$ with $a, b \geq 0$ and coverage exactly $1 - \alpha$, show that the expected length is smallest when $a = b = t_{n-r, \alpha/2}$. (Use that the t density is symmetric and strictly decreasing in $|x|$.)

Solution. Coverage is $F(b) - F(-a)$, with F the $t(n-r)$ distribution function, and the expected length is proportional to $a + b$. If $a < b$, shifting $[-a, b]$ slightly left gains density $f(a)$ and loses $f(b) < f(a)$, raising the coverage, so a strictly shorter interval has coverage $1 - \alpha$. Hence $a = b$.

Exercise C2.

Intervals for σ^2 of the form $[\text{SSE}/c_2, \text{SSE}/c_1]$ with $\Pr(c_1 \leq \chi^2(\nu) \leq c_2) = 1 - \alpha$ all have exact coverage. Show that the one with the smallest expected length satisfies

$c_1^2 f(c_1) = c_2^2 f(c_2)$, where f is the $\chi^2(\nu)$ density, and that the equal-tailed choice of [Proposition 12.1.4](#) does not satisfy this condition.